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International Institute of Information Technology, Hyderabad
(Deemed to be University)

Introduction to Brain and Cognition (CS9.311) Spring-2026
Question cum Answer Booklet
Quiz (Set C)

Max. Time: 45 Minutes

31-03-2026

Max. Marks: 20

Roll No: 2025121009

Programme: LCP

Invigilator's Signature:

Special Instructions about the exam

1. This is a closed book exam and all questions are compulsory.
2. There are 13 objective questions and 2 descriptive questions for a total of 20 marks.
3. Only **one correct** answer per MCQ unless stated otherwise.
4. MSQ (Select All That Apply) — 0 marks if not all correct options are selected.
5. In case of any doubt, please answer while listing your assumptions appropriately.

General Instructions to the students

1. Place your Permanent / Temporary Student ID card on the desk during the examination for verification by the Invigilator.
2. Use of calculators is not allowed. Reading material such as books (unless open book exam) are not allowed inside the examination hall.
3. Borrowing writing material or calculators from other students in the examination hall is prohibited.
4. If any student is found indulging in malpractice or copying in the examination hall, the student will be given 'F' grade for the course and may be debarred from writing other examinations.

Good Luck!

25 P1

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Section A: Objective Questions (15 Marks)

MCQ: 1 mark each | MSQ: 2 marks — 0 marks if not all correct options are selected

- Q1. A person looks at a series of flashing lights arranged in a circle and perceives a single moving light instead of separate flashes. This phenomenon best illustrates:
- ☒ A) Gestalt perception of wholes
 - ☐ B) Operant conditioning
 - ☐ C) Classical conditioning
 - ☐ D) Insight learning
- Q2. The deep groove on the brain surface that separates the frontal lobe from the parietal lobe is called the:
- ☒ A) Central sulcus
 - ☐ B) Longitudinal fissure ← left and right
 - ☐ C) Transverse fissure ← Cerebrum/cerebellum
 - ☐ D) Sylvian fissure ← frontal & parietal separating temporal
- Q3. A rat learns to press a lever for food. A strict behaviourist would explain this as:
- ☐ A) Perceptual reorganization
 - ☐ B) Formation of a cognitive map ← later by Tolmann (1950s-60s)
 - ☒ C) Reinforcement strengthening a response
 - ☐ D) Insight into the environment
- Q4. According to the principle of localisation of function, the occipital lobe is primarily associated with:
- ☐ A) Touch processing
 - ☐ B) Motor planning ← Frontal
 - ☒ C) Visual processing
 - ☐ D) Auditory processing ← Temporal
- Q5. In the Motor System, which brain area is specifically associated with the coordination of complex motor patterns?
- ☐ A) Primary Motor Cortex (M1) ← Plan and Execute
 - ☐ B) Prefrontal Cortex
 - ☒ C) Cerebellum
 - ☐ D) Basal Ganglia ← Voluntary, Reinforcement Learning
- Q6. Which neuroimaging method is primarily used to study the metabolic function of the brain by tracking a radioactive tracer?
- ☐ A) EEG
 - ☒ B) PET ← Tomography → Invasive, records Glucose (photon emission)
 - ☐ C) MRI
 - ☐ D) fNIRS
- Q7. Which psychological movement rejected introspection and insisted that psychology should focus only on mechanisms and processes that are objective and observable?
- ☒ A) Behaviourism ← John B. Watson (1900)
 - ☐ B) Gestalt Psychology
 - ☐ C) Structuralism
 - ☐ D) Cognitivism
- Q8. Which of the following best distinguishes perception from sensation?
- ☐ A) There is no meaningful distinction between sensation and perception
 - ☐ B) Sensation involves interpretation; perception involves raw data collection
 - ☒ C) Sensation is the gathering of raw sensory data; perception is the active interpretation of that data
 - ☐ D) Sensation is the brain's active interpretation; perception is the raw input from receptors

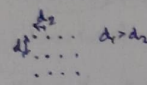
Q9. A Capgras patient can correctly identify a familiar person but reports that the person "feels unfamiliar" and is therefore believed to be an imposter. This pattern most strongly suggests:

(Select ALL that apply — 2 marks, 0 if incomplete)

- ☒ A) A functional disconnection exists between face recognition areas and limbic structures (FFA) (amygdala)
- ☒ B) The affective component of recognition contributes to beliefs about identity
- ☒ C) The ventral visual pathway supporting face recognition is intact
- D) Familiarity judgments depend solely on perceptual identification
- E) The deficit reflects a generalized memory impairment
- ☒ F) There is a disruption in the autonomic/emotional response to familiar stimuli

✓ 400d

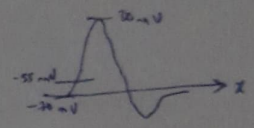
Q10. Which Gestalt principle describes the tendency to perceive objects that are close to each other as forming a group?

- ☒ A) Proximity 
- B) Closure
- C) Continuity
- D) Similarity

✓

Q11. According to the action potential mechanism, what is the typical resting membrane potential of a neuron?

- A) 0 mV
- B) +40 mV
- C) -10 mV
- ☒ D) -70 mV ← -55 mV is the threshold and can go up to 30 mV while depolarization
-70 mV is the resting potential



Q12. A 55-year-old patient can initiate movements but shows difficulty in tasks like buttoning a shirt or writing. Movements lack smoothness and precision during complex motor activities. Which brain area is most likely affected?

- A) Cerebellum ← Coordination
- B) Basal ganglia ← Voluntary, RL
- C) Prefrontal cortex
- ☒ D) Primary motor cortex (M1) ← Execution i.e. writing / buttoning

✗

Q13. In the hierarchical organization of the brain, which structure is part of the "Reptilian complex" (Triune brain theory) responsible for basic physiological functions?

- A) Limbic System ← Paleomammalian complex
- B) Amygdala ← Emotion, fear
- C) Neocortex ← Neomammalian complex
- ☒ D) Cerebellum (Core structures)

✓

Q14. Which brain imaging technique measures metabolic function and is classified as invasive?

- ☒ A) PET ← Positron Emission Tomography - Measures Glucose
- B) EEG ← Non-invasive, electrodes on scalp
- C) TMS
- D) fMRI ← Non-invasive, blood oxygenation level dependent measuring

✓

Section B: Short Answer Questions (5 Marks)

(2.5 marks each)

Q1. Distinguish between open-loop and closed-loop motor control. Give one example of each.

Open-loop motor control is a scheme having no-feedback loop, i.e. some ~~action~~ ^{action} that cannot be corrected if there is any error mid-way.

e.g. Throwing a dart ~~to~~ Cannot be corrected mid-flight.

State → Decide → Execute → Predict → Observe → Correct

Closed-loop motor control is a scheme having a feedback loop, i.e.

Some action that can be corrected if there is any error mid-way.

e.g. Driving a Car, Balancing on a rope.

State → Decide → Execute → Predict → Observe → Correct

When we ~~are~~ tend to lose balance, we fix our posture, similarly while driving initially consciousness → Cortical → Subcortical ← final stage

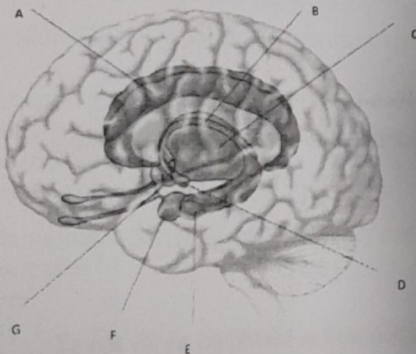
↓
Fully aware
closed-loop

↓ After expertise,

open-loop, but still if anything new comes, again closed loop (e.g. construction)

(Habit)

Q2. Name any four subcortical structures visible in the diagram below and briefly state the role of any one of them.



C: Thalamus ✓

E: Hippocampus ✓

F: Amygdala ✓

G: Hypothalamus ✓

Thalamus (C): Input Gateway for sensory actions

It gets all the task information and sends to respective parts of the brain, according to which part performs what.

Hippocampus (E): Responsible for Long-Term Memory

Amygdala (F): Responsible for Emotions and Fear stimuli

Is seen damaged in Capgras Syndrome Patients

Hypothalamus (G): Responsible for maintaining and stabilizing:

- (1) Body Temperature
- (2) Heart beat Rhythm
- (3) Breathing regulation
- (4) Sleep-Wake Cycle

Also called the mid-brain (or) brainstem.

